Process Cube for Software Defect Resolution

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***Abstract*—Online Analytical Processing (OLAP) cube is a multi-dimensional dataset used for analyzing data in a Data Warehouse (DW) for the purpose of extracting actionable intelligence. Process mining consists of analyzing event log data produced from Process Aware Information Systems (PAIS) for the purpose of discovering and improving business processes. Process cube is a concept which falls at the intersection of OLAP cube and process mining. Process cube facilitates process mining from multiple-dimensions and enables comparison of process mining results across various dimensions. We present an application of process cube to software defect resolution process to analyze and compare process data from a multi-dimensional perspective. We present a framework, a novel perspective to mine software repositories using process cube. Each cell of process cube is defined by metrics from multiple process mining perspectives like control flow, time, conformance and organizational perspective. We conduct a case-study on Google Chromium project data in which the software defect resolution process spans three software repositories: Issue Tracking System (ITS), Peer Code Review System (PCR) and Version Control System (VCS). We define process cube with 9 dimensions as issue report timestamp, priority, state, closed status, OS, component, bug type, reporter and owner. We define hierarchies along various dimensions and cluster members to handle sparsity. We apply OLAP cube operations such as slice, dice, roll-up and drill-down, and create materialized sublog for each cell. We demonstrate the solution approach by discovering process map and compare process mining results from control flow and time perspective for Performance and Security issues.**

Keywords*—Empirical Software Engineering, Issue Tracking System, Mining Software Repositories, OLAP, Peer Code Review System, Process Cube, Process Mining, Version Control System.

I. Research Motivation and Aim

Stand-alone process analysis is the common way of analyzing processes in today's process mining approaches [1]. There may be multiple variants of same process influenced by multiple factors like different working patterns, diverse business requirements and change with time (concept drift). Therefore, it is of great importance to cluster cases based on various attributes and compare process mining results [2]. This enables identification of more efficient paths and practices, thus, other process variants can be modified to improve performance. Process cube is a multidimensional data structure where events and process models are organized using different dimensions to facilitate comparison of multiple processes [1]. Process Cube View (PCV) is defined using Process Cube Structure (PCS) and Event Base (EB). PCS defines the dimensions of the cube, and EB is the complete set of attributes used for dimension mapping and materialized sublog creation. Relevant event attributes are selected as dimensions based on the type of comparisons to be performed. Since the idea of process cube is related to OLAP, we can exploit its features and perform functions like slice, dice, roll-up and drill-down.

ITSs (like Bugzilla[1] and Mantis[2]), PCRs (like Gerrit[3] and Rietveld[4]) and VCSs (like SVN[5] and Mercurial[6]), are workflow management systems jointly supporting the bug reporting and resolution process in software maintenance. Free-Libre/Open Source Software (FLOSS) projects like Google Android and Chromium follow a process consisting of issue reporting in ITS followed by patch submission for review in PCR (if required, to resolve reported issue) and committing source code change using VCS [3]. An issue reported in ITS is characterized by many attributes like *Priority, Type* of issue, *Component* to which issue pertains and *OS*. A common issue resolution process is defined irrespective of these attributes. This work is motivated by the need to compare and understand the differences in process for cases with different characteristics along various dimensions. We believe it is important to consider how differences along various dimensions produce process variants. Specific research aims of the work presented in this paper are as following:

1) To define PCV for software defect resolution process.
2) To investigate applications of OLAP operations like slice, dice, roll-up and drill-down on process cube, thus, compare process along various dimensions.
3) To discover process model using state-of-the-art algorithms for each process cell. Also evaluate and compare metrics from multiple perspectives like control flow, time (bottleneck), and organizational to identify inefficiencies and suggest modifications for improved performance.
4) To conduct a case-study on a popular FLOSS project, Google Chromium to illustrate the usefulness of proposed approach and compare process mining results across various process variants.

II. Related Work and Research Contributions

In this section, we discuss closely related works and present the novel research contributions of the study presented in this paper in context to the existing work. We organize closely related work into following three lines of research:

A. Data Warehousing Environment in Software Development

Ruiz *et al.* present a DW environment to support the implementation of a process measurement and analysis program in a CMM Level 2 organization [4]. DW provides a centralized and

[1] http://www.bugzilla.org/
[2] http://www.mantisbt.org/
[3] http://code.google.com/p/gerrit/
[4] http://code.google.com/p/rietveld/
[5] http://subversion.apache.org/
[6] http://mercurial.selenic.com/

unified view of all projects and stores project-related data to support monitoring of software development according to the defined metrics [4]. Colaço *et al.* propose a data warehousing approach for software development data analysis [5]. They present a dimensional model for software change history and analyze its data using two tools: a change history miner and a corrective maintenance dashboard [5]. It is possible and viable to keep a DW focused on code construction [5]. Silveira *et al.* present a software process data warehousing architecture, SPDW+ as a solution to frequent, seamless, and automated capturing of software quality metrics, and their integration in a central repository for full range of analysis [6].

B. Process Mining Software Repositories

Gupta *et al.* present an application of integrating and process mining three software repositories (ITS, PCR and VCS) from control flow and organizational perspective [3]. They propose a framework to discover runtime process model for bug resolution process, identify bottlenecks, define and detect basic and composite anti-patterns and discover metrics such as handover of work, subcontracting, joint cases and joint activities [3]. Kim *et al.* propose a distributed workflow mining approach to discover workflow process model incrementally amalgamating a series of vertically or horizontally fragmented temporal work-cases [7]. Poncin *et al.* present a framework called as FRASR (FRamework for Analyzing Software Repositories) that facilitates combining and matching of events across multiple repositories like mail archives, subversion and bug repositories followed by assignment of role to each developer [8]. Song *et al.* apply process mining technology to common event logs obtained from five information systems for behavior pattern mining [9].

C. Process Cubes

Aalast *et al.* formalize the notion of process cubes where events and process models are organized using different dimensions which allows interactive analysis and exploration of process [1]. Mamaliga *et al.* developed an initial prototype for process cubes, ProCube using the process mining framework ProM and the Palo OLAP toolset to allow comparison of multiple processes [2]. ProCube plugin in ProM creates sublogs per cell on-the-fly (unlike traditional process mining approaches) and visualizes the discovered process models, social networks and dotted charts [2].

In context to the existing work, we present the following novel and unique research contributions in this paper:

1) While there has been work done in the area of data warehousing for software development, to the best of our knowledge, the study presented in this paper is first work on application of OLAP model to *define process cube notion for software defect resolution*.
2) While stand-alone defect resolution process has been process mined from multiple perspectives, this paper presents a novel framework to *facilitate comparison between various process variants by applying OLAP operations on process cube*.
3) We conduct an *in-depth empirical analysis on Google Chromium project (open-source) data extracted from Google issue tracker, Rietveld PCR and Subversion VCS* to demonstrate the application and effectiveness of our approach. We compare process mining results for different issue *Type* such as Security and Performance. We present results for runtime process discovery, activity analysis, transition analysis and bottleneck identification.

III. Research Methodology and Framework

We conduct experiments on dataset downloaded from Google ITS, Rietveld PCR and Subversion VCS of Google Chromium browser project. It is a large, long-lived and complex open source software project. Issue reports, patches and commit details for Google Chromium browser are publicly available. Hence, the experimental analysis presented in this paper can be replicated, used for benchmarking and comparison by other researchers. Fig. 1 depicts the multi-step framework adopted for this research work. It has majorly three steps: 1. *data extraction, integration and transformation*, 2. *process cube view and operations*, and 3. *process mining* (process cell metrics). During the first step, we extract data for complete lifecycle of an issue reported in ITS using JSON-RPC and XML-RPC. We notice that for some cases, lifecycle spans across multiple software repositories (ITS, PCR and VCS). Therefore, the data is extracted and integrated from multiple information systems (IS). As shown in Fig. 1, the complete transformed Event Base (EB) having trace data for all cases is stored in MySQL relational database. PCS is defined and OLAP operations are performed to obtain and compare process mining results. Materialized sublog derived from EB for each cube cell can be process mined from multiple perspectives

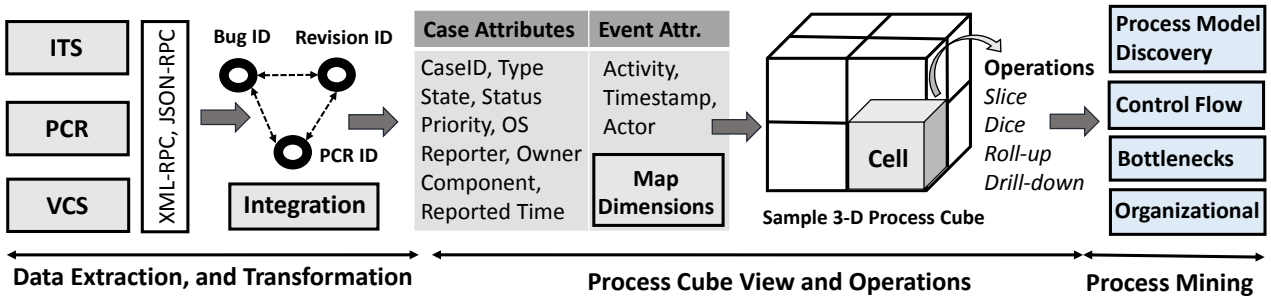


Fig. 1: Research framework: multistep framework with majorly three stages: 1. Data Extraction, Integration and Transformation, 2. Process Cube View and Operations, and 3. Process Mining.

TABLE I: Experimental Dataset Details (Chromium Project).

Attribute	Value
First ITS issue creation date	1 Jan 2009
Last ITS issue creation date	31 Dec 2012
Total extracted ITS issues	177926
Total closed ITS issues	145326 (81.67% of extracted)
Total closed ITS issues with lifecycle spanning 3 IS	39550 (27.21% of closed)
Total closed ITS issues with type *Security*	2271 (1.56% of closed)
Total closed ITS issues with type *Performance*	716 (0.49% of closed)

like control flow, conformance, time and organizational using various measures [10].

A. Data Extraction, Integration and Transformation

We extract 4 year data of Google ITS starting from 1 January 2009 to 31 December 2012 using JSON-RPC (refer Table I). Data for 177926 issue reports is extracted and we find that around 82% issues are closed. As shown in Table I, 39550 closed issues require source code change (patch) for resolution which are peer reviewed in PCR to avoid defects before they are committed into source code, VCS. Eventually, these cases have lifecycle spanning three IS which are not explicitly linked. Therefore, integration becomes very important and performed using the text analysis approach proposed by Gupta *et al.* [3]. We obtain a complete EB, that is, trace for all the cases with two category of attributes:
1. **Case Attributes:** Properties associated with a case which remain constant for all the events pertaining to same case. For example: *CaseID, Type, State, Closed Status, Priority, OS, Reporter, Owner, Component and Reported Time* of an issue.
2. **Event Attributes:** Properties characterizing each event in lifecycle of an issue like *Activity, Actor* (resource) *and Timestamp* (when the activity is performed).

Table III displays list of all the extracted case and event properties (attributes) with description. Some case properties are not defined for all the cases, hence, NULL is assigned as value. For instance, many cases do not have a priority label associated with them, thus, we assume priority as NULL. For event attributes, we identify a set of activities to embark progress of an issue towards resolution as shown in Table II. There are 8 activities from ITS. We group all the open status labels[7] (like *unconfirmed, untriaged, available, assigned and started*) and represent as *I_Open* state because these labels do not mark significant difference in progress. If a patch is submitted to PCR system, we denote it as state *C_Creation* and once the patch review is done, we represent it as *C_reviewed*. Patch may or may not be committed to source code depending on approval of the reviewers. If a patch is approved after all the corrections, we indicate commit to VCS as *V_commit* (refer Table II). We capture corresponding actor and ts for all the activities and store in EB. In the next steps, we define PCS using the properties in EB as universal set of dimensions, D. Also we derive materialized sublog from the EB for each process cell to perform process mining.

[7]http://www.chromium.org/for-testers/bug-reporting-guidelines

TABLE II: List of Activities, its Significance.

Activity	Description
Issue Tracking System Activities	
I_Creation	Issue reported in ITS
I_Open	Open bug status label
Fixed	Resolved as Fixed
Invalid	Illegible, spam etc.
Duplicate	Issue has been reported in another bug, or shares the same root cause as another bug.
WontFix	Can't reproduce, Working as intended, Obsolete
Verified	Resolution verified
I_Closed	ITS progress ends
Code Review System Activities	
C_Creation	Patch reported in PCR
C_Reviewed	Code review process ends
Version Control System Activities	
V_Commit	Code change committed

B. Process Cube View

Process cube links the process mining framework to the existing OLAP technology. It is a multidimensional structure built from event log data for efficient process mining analysis and comparison [1]. Unlike OLAP where each cell is a numerical measure, process cube cell consists of a set of events derived from EB based on dimension values. We identify dimensions for defect resolution PCS. All the properties, D stored in EB can be selected as dimension, D_{sel} for process cube. However, this may lead to inefficient performance due to sparsity. Sparsity is a challenge because of dimensions with very large domain leading to highly infrequent occurrence of its values in complete EB. This results in missing event sublog for majority of the process cells after applying various OLAP filtration operations, consequently, degrades the performance. We can handle sparsity by following ways:

- **Dimension Subset Selection:** Dimensions of final process cube can be a subset of all the characterization properties stored in EB, that is, $D_{sel} \subseteq D$. However, it is not always a viable solution to remove attribute causing sparsity from cube dimension because it involves many storage and performance challenges [2].
- **Hierarchies:** With an increase in granularity of attribute value, frequency of value decreases in complete log. Accordingly, we can use values at higher level to reduce the number of distinct members per dimension. This can only be applied to dimensions with hierarchical domain. For example, from Table III, *Priority*[8] is a dimension with hierarchy *Class* $\rightarrow$ *Value* where *Class* can be High or Low. High can have value 0 or 1 while Low can have value 2 or 3. Here we can roll up to higher level (*Class)* to handle sparsity.
- **Clustering:** We can group multiple values using well defined grouping criteria. Thus, reducing the number of distinct members without compromising the quality of analysis. For example, domain of *Owner* is very large. Therefore, we can create classes based on the duration of participation and productivity. As shown in Table III, we reduce domain of *Owner* dimension to

[8]http://www.chromium.org/for-testers/bug-reporting-guidelines/chromium-bug-labels

TABLE III: Case and Event Attributes in Event Base with the Description. D/ND Denotes if the Attribute is a Dimension (D) or Not Dimension (ND) of Process Cube. Domain is a Set and Type of Values Possible for Each Attribute.

Attribute	Description	D/ND	Sparsity	Domain
Case Attributes : Common for all events belonging to same case				
CaseID	Unique ITS issue ID	ND	–	Integer
Type [11]	Issue type	D	No	Regression, Security, Crash, Performance, Usability, Polish, Cleanup
State	Issue is in process or closed	D	No	Open, Closed
Closed Status[7]	Final issue resolution label	D	No	Fixed, Verified, Duplicate, WontFix, Invalid, External Dependency, FixUnreleased, Icebox
Priority	Importance of fixing an issue	D	Hierarchy	Class (High or Low)→Value (0 and 1; 2 and 3)
OS	Operating System to which issue pertains	D	No	All, Chrome, Linux, Mac, Windows
Reporter	Person who reports issue	D	Clustering	Low, Medium, High : Based on frequency of issues reported
Component[8]	Component to which issue pertains	D	Clustering	Content, Internals, Platform, UI, OS: Abstract to broad category
Reported Time	Published time for issue	D	Hierarchy	Year→Month→Day→Time
Owner	Person who manages issue	D	Clustering	Long Term, Short Term Contributors: Duration and productivity based [12]
Event Attributes : Different for each event of a case				
Activity	Task performed	ND	–	Activities captured for issue resolution progression [3]
Timestamp	Time when activity is performed	ND	–	Datetime
Actor	Person who performs activity	ND	–	People participating in resolution process

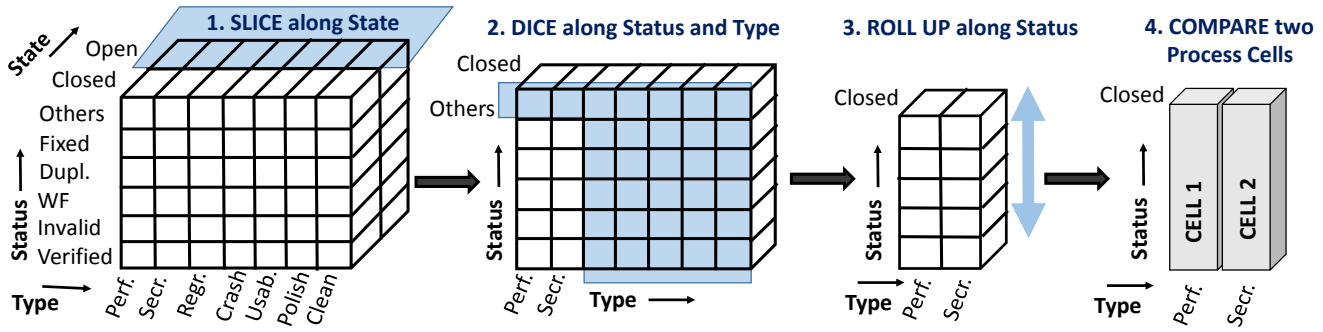


Fig. 2: Illustrate set of OLAP operations on sample 3-D process cube to compare process mining results for *Performance* and *Security* type issue resolution.

2 members: Long Term Contributors (LTC) and Short Term Contributors (STC) [12]. This allows for delta analysis between the issue resolution process adopted by LTC vs STC. Similarly we can classify values for other sparse dimensions like *Component* and *Reporter* according to business analysis requirements.

Based on our analysis, we conclude that it is a technical challenge to efficiently handle sparsity and select dimensions for a process cube. As observed in Table III, all the case attributes are selected as cube dimension except *CaseID* because it is unique so highly sparse. Also we believe that filtering based on *CaseID* is not very interesting for business analysis. In Table III, we present ways to handle sparse dimensions and domain for each attribute. Here we identify 7 members for dimension issue *Type* such as regression, security, performance, crash, cleanup, polish and usability. [11]. We notice that no event attribute is used as process cube dimension though we derive sublog with four fields: *CaseID, Activity, Timestamp* and *Actor* using event attributes from the all-encompassing EB. Step of extracting an event log from EB for process cube cell is known as *materialization step*. During this step, if value for case attributes belongs to the permitted set of values along filtration dimensions then we add case log to materialized sublog for a process cell. We pass derived materialized sublog as input to standard process mining tools like Disco[9](commercial) and ProM[10](open source) for process discovery. Also other process mining measures of interest are evaluated and compared systematically for all the cells.

C. Process Cube Operations

Process cube idea is similar to OLAP thereby, we explore application of following OLAP operations to perform diverse analysis [1]. We illustrate application of these operations on sample process cube for software defect resolution. We consider an instance cube with three dimensions: *1. Type, 2. State, and 3. Closed Status*. For remaining case attributes in event base, all the domain values are allowed and no filtration constraint is applied on them. The objective is to compare process derived from execution data of different bug *Type* resolution.

1) *Slice*: Selects single value for one of the dimension, thus, reducing total number of dimensions by one. In Step 1 of Fig. 2 we slice process cube along *State* dimension and obtain a subcube with value for *State* as closed, that is, only closed issues are considered for analysis.
2) *Dice*: Defines a subcube by performing selection of single or multiple dimensions thereby, reducing number of members along selected dimensions. We obtain a subcube in Step 2 of Fig. 2 by selecting

[9]http://www.fluxicon.com/

[10]http://www.promtools.org/prom6/

Type as Performance and Security, and *Status* as Fixed, Invalid, Duplicate, WontFix and Verified. All the cases with final status (like External Dependency, FixUnreleased and Icebox grouped as Others for the sake of presentation) apart from these 5 are filtered.

3) *Roll Up*: Summarizes data along a dimension. For example, as depicted in Step 3 of Fig. 2 we roll up and combine all the selected *Closed Status* values (from Step 2). As a result, no partitioning based on final *Closed Status*. This reduces number of cells along the rolled up dimension.
4) *Drill Down*: Represents data at more specialized level of hierarchy with more number of members along the dimension. It is opposite of roll up.

Eventually, after performing sequence of above operations on sample process cube, we compare process variants for issue *Type*, Performance and Security. There has been research work done highlighting the importance of considering different types of bugs in software quality research and practice [11] [13]. A common issue resolution process is defined for Google Chromium project, here we aim to bring out the differences in process in-practice for different bug types. We obtain materialized sublog from complete 4 years event log data. Event log for all the cases with bug *Type* as Performance or Security having final *Status* as Fixed, Invalid, Duplicate, WontFix or Verified is added to corresponding materialized sublog for Cell 1 and Cell 2 respectively. As a result, we obtain two logs for process mining and compare from various perspectives like control flow and time using metrics proposed in following sections. We make the dataset used in our study publicly available at github[11] so that our experiments can be replicated and used for benchmark purposes by other researchers.

IV. Control Flow Perspective

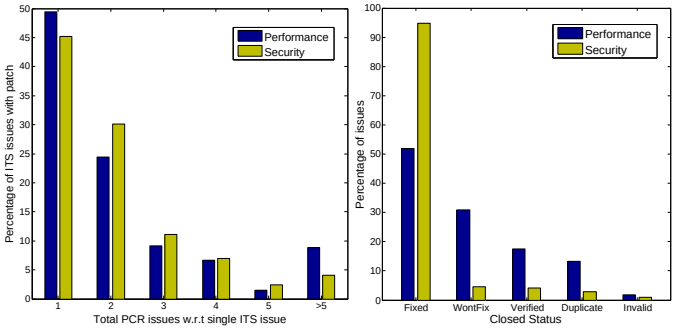
Control flow perspective helps to identify the activities performed and the order of their execution. We believe it is interesting to study automatically obtained process model from the materialized sublog of Performance and Security type bugs. It reflects the reality and can be used as input and actionable information to accommodate changes for resolution process of different bug types. For each cell, we import preprocessed sublog generated during the progression of an issue into Disco to discover runtime process model and other statistical information. Disco miner is based on the proven framework of Fuzzy Miner with completely new set of process metrics and modeling strategies[12]. We select ITS issue ID as *CaseID* (for process map generation in Disco) to associate all activities pertaining to same issue ID. As a result, visualize the complete life cycle of a bug and control transfer between different systems.

We have a log of 704 and 2271 cases corresponding to Cell 1 (Performance) and Cell 2 (Security) respectively of process cube in Fig. 2. There are total 11 states in the process map where 8 states are for ITS, and 3 for PCR and VCS. Fig. 3a and Fig. 3b depicts process map for issues with *Type* as Performance and Security respectively. For the sake of simplicity and clarity, only the core transitions are presented in process map because the model with all infrequent transitions is complex and looks like spaghetti. Label on edges represents absolute frequency of transition and value in an activity node is total number of times that activity is performed in complete event log. Shade of a node and thickness of an edge in process map corresponds to absolute frequency of activity and transition respectively. Dark shade is for comparatively higher frequency similarly, thick edge corresponds to more frequent transition. We perform statistical analysis of derived process map for activity frequency, transitions and commonalities.

A. Activity Frequency Analysis

Patch Creation: Patch is submitted to PCR for review in order to resolve an issue. We identify that a significant number of Performance issues, that is, 328 (46.59%) have atleast one patch submitted to PCR while 736 (32.40%) of Security issues have patch(es) to resolve an issue. For both the *Types*, majority of the cases have 1 or 2 PCR issues in their lifecycle as clearly shown in Fig. 4a. However, there are outliers with more than 5 PCR issues to resolve same ITS issue. From Fig. 3a and 3b, the total number (absolute frequency) of PCR issues created (activity *C_Creation*) for *Type* Performance and Security are 741 and 1539 respectively. Out of these PCR issues, 674 (90.95%) and 1473 (95.71%) are committed for Performance and Security respectively. Hence, we can say that comparatively more Performance issues require patches for resolution when in fact, the patches submitted for Security issues are more likely to get committed. Overall the quality of submitted patches is good as the percentage of patches getting committed is quite high.

Closed Status: From Fig. 4b, we notice that there is a significant difference in closed status distribution between two issue types. Interestingly, around 95% of the Security issues are getting resolved as *Fixed* which is quite high and reflects effective bug resolution. Whereas, only 50% Performance issues are *Fixed*. A good number of Performance issues are marked as *Duplicate* or *WontFix* which means that either the reported issues do not have sufficient details, hence, not reproducible or are not relevant. Very less Security issues are marked as *Duplicate* or *WontFix* which highlights the importance given to Security issues. Therefore, efforts should

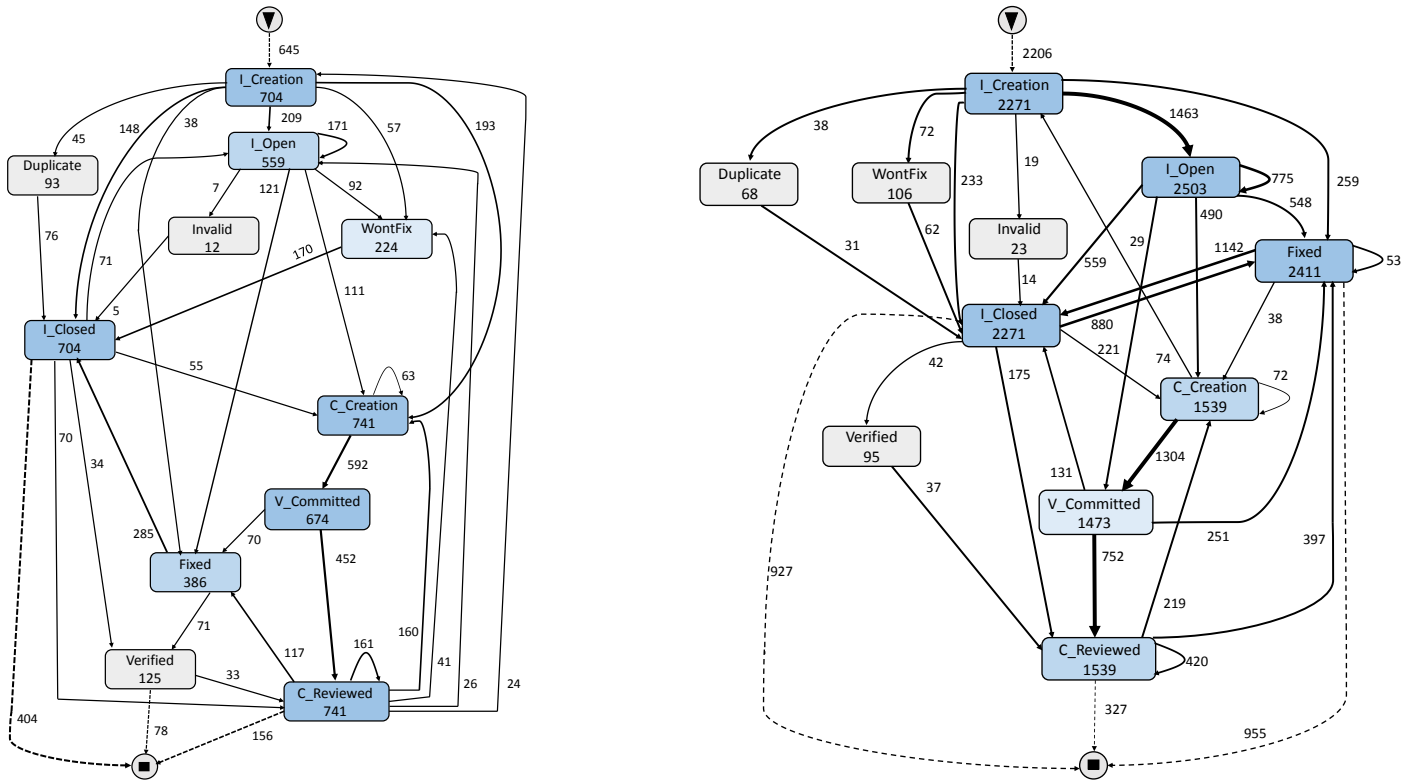


(a) Percentage of total ITS issues with total PCR issues in lifecycle.

(b) Closed status distribution for Performance and Security issues.

Fig. 4: Activity frequency analysis

[11] https://github.com/Mining-multiple-repos-data/ProcessCubeData
[12] http://fluxicon.com/disco/files/Disco-Tour.pdf



(a) Process model for Performance issues

(b) Process model for Security issues

Fig. 3: Process model to compare Performance and Security Type issue resolution. Edges and nodes are labeled with absolute frequency, thickness of edge and shade of node corresponds to absolute frequency.

be made to fix more Performance issues and improve the quality of resolution for them.

According to Google Chromium bug resolution guidelines[7], it is recommended that a fix is *Verified* (means that the fix has been tested and confirmed). However, we notice that very less percentage of issues are verified for their resolution. It is even lesser for Security issues (5%) as compared to Performance issues (around 18%) signaling the need to make more efforts towards issue verification.

B. Transition Analysis

Unique Traces: Ordered sequence of activities performed during the lifecycle of a case is known as trace. We observe fairly large number of trace variants where we consider two variants unique even if one transition is different. We have total number of variants as 279 and 549 for Performance and Security type issues respectively. The distribution is skewed, as a result, top 5% traces depict 50% and 67% of cases for Performance and Security issues respectively. We find that for both the types, most common traces have lifecylce limited to ITS only, and not spanning to PCR and VCS. It indicates that the traces spanning across all the 3 IS have more variants with less frequency because majority cases have one while some have multiple PCR issues in their lifecycle. These multiple PCR issue have different ordering of *C_Creation* and *V_Commit*.

Events in Lifecycle: Total number of events in lifecycle ranges from 3 to as high as 46 and 38 for Performance and Security issues respectively. We observe that the resolution process is quite stable as majority of the cases have less than 10 events with some outliers requiring more activities for resolution. General trace with 3 events can be represented as: *I_Creation* → *Closed Status* → *I_Closed*,
while the general trace with 4 activities: *I_Creation* → *I_Open* → *Closed Status* → *I_Closed*

Control Transfer between 3 IS: As evident from Fig. 3a and 3b, majority of the cases (645 for Performance and 2206 for Security) start their lifecycle from ITS with activity, *I_Creation*. For cases with process spanning across all the 3 IS, control transfer between three systems in most of the cases is as *ITS* → *PCR* → *VCS*, that is, lifecycle starts from ITS. According to Google Chromium bug resolution guidelines[13], it is recommended to first report an issue in ITS followed by patch submission to PCR for resolution thereafter, if approved gets committed to VCS. However, we identify the cases where this practice is not followed and lifecycle begins with patch submission to PCR (*C_Creation*). 18% (59 cases) of total Performance issues with resolution spanning 3 IS start their lifecycle from PCR system. Unlike Performance issues, only 8.42% (62) of Security cases have lifecycle starting from PCR system with delayed issue reporting in ITS to maintain

[13]http://dev.chromium.org/developers/contributing-code

a record of code changes with respect to issue to be addressed.

Anti-patterns: We identify undesired, erroneous transitions and cause of their existence in our process model. It is important for a process analyst to improve overall quality by minimizing occurrence of such transitions. We notice from derived runtime process model (refer Fig. 3a and 3b) that *I_Closed* is end state for comparatively less percentage of cases, that is, 57.39% and 40.82% for Performance and Security respectively. Ideally it should be end state for all the cases. However, we observe outgoing edges from *I_Closed* towards other states like *I_Open*, *Fixed*, *Duplicate*, *WontFix*, *C_Creation* and *C_Reviewed*. Potential reasons for outgoing edge to the following states are as follows:

- *I_Open, Fixed, Duplicate, WontFix:* Issue is closed but if the resolution deemed incorrect, it is reopened. Such cases increase maintenance cost, degrade overall user perceived quality of the bug fixing process and leads to unnecessary rework by busy practitioners [14]. We find that there are total 133 and 892 cases with transitions from *I_Closed* to other states of ITS for Performance and Security issues respectively, which signifies that the issue was reopened and resolved. Interestingly, out of 892 cases of Security issues, 880 transitions are to state *Fixed* which means that either additional information becomes available or the root cause is understood later. This leads to fixing of otherwise closed bugs. Also an issue is reopened if a bug regresses. Overall, we can say that more Security issues (around 39.27%) are reopened and preferably *Fixed*. It is good from the perspective that maximum Security issues are *Fixed*, at the same time, reopening of many issues indicate inefficiency and delay in resolution process. Therefore, process owner should make efforts to fully understand an issue before marking it closed to minimize chances of reopen.

- *C_Creation, V_Committed, C_Reviewed:* For Performance and Security issues, there are 133 and 410 direct transitions respectively to PCR and VCS states indicating that the submitted patch review is in progress while the issue gets closed.

Signature Patterns: Set of transitions occurring commonly in both the process models and are interesting to study as part of issue resolution lifecycle are marked as signature patterns. From Fig. 3a and 3b, we observe a clear cycle as *C_Creation* $\rightarrow$ *V_Committed* $\rightarrow$ *C_Reviewed* $\rightarrow$ *C_Creation*, which means that for majority of the cases with multiple PCR issues, the PCR issues are created sequentially. It is because if the committed patch is not sufficient to address an issue only then the need for more code changes is realized thereby, followed by new patch report in PCR. For both the processes, we also observe self loop in *C_Creation* indicating that subsequent patches are submitted to PCR even before the review of previous patch. However, this is happening for comparatively less cases (63 and 72 for Performance and Security issues respectively) as it is not easy to identify the next files to be manipulated without committing already suggested changes.

V. Time Perspective

Gupta *et al.* identify specific part within the runtime process map which is relatively time consuming and reduces overall performance of the end-to-end process, that is, bottleneck for stand-alone process [15]. We propose metric to identify bottlenecks using multiple process variants to help process analyst improve the performance. We observe that some transitions are very time consuming in a process irrespective of other variants (*General bottleneck*) while there exists transitions which are inefficient in some variants when compared with others (that is, *Exclusive bottleneck*).

A. Exclusive Bottleneck

Some transitions are present in all variants. Still mean time for the same transition varies by a factor of 2 or more across the variants. It implies that same activity is performed more efficiently in some variants highlighting the scope of improvement in others. We propose Bottleneck Ratio (*BNR*) to identify such instances:

$$\boxed{BNR = \frac{|(P_t - P'_t)|}{min(P_t, P'_t)}}$$

where P_t = Mean time for transition *t* in process 1,
and P'_t = Mean time for transition *t* in process 2

BNR measures the extent of difference with respect to minimum time taken for the same transition. If *BNR* ≥ 1, it means that time taken by a process is atleast double of minimum time for the transition. We suggest use of *BNR* to compare two processes while if the number of process variants is more than two then *Adjusted Box Plot* can be used [16]. Outlier in box plot corresponds to the bottleneck, hence, need for improvement.

Here, we identify bottleneck (using *BNR*) present in core transitions of two process variants presented in Fig. 3a and 3b. Some of the examples are listed in Table IV where we find that *BNR* ratio is quite high for some of the transitions.

1) As observed in Table IV, identification of *Duplicate* and *WontFix* issues is very inefficient for Performance issues with *BNR* greater than 15. Also from Fig. 4b, percentage of such issues is more for Performance type. Thus, there is clear need to address the cause of delay in determining such issues to improve the quality of Performance issue resolution process.
2) Fig. 3a and 3b reveal that in comparison to Performance issues, Security issues have more open status label in their lifecycle (around 70%), that is, activity *I_Open* is more frequent in Security issues as

TABLE IV: Transition with BNR where P_t Corresponds to Performance and P'_t to Security. Minimum of Two is Represented in Italics. (h: hour, d: day, w: week)

Transition	P_t	P'_t	**BNR**
I_Creation $\rightarrow$ I_Open	17.2 d	*4.9 d*	2.51
I_Creation $\rightarrow$ Duplicate	37.5 d	*50.5 h*	16.82
I_Creation $\rightarrow$ WontFix	24.2 w	*8.4 d*	19.16
V_Committed $\rightarrow$ C_Reviewed	*6.7 d*	36.1 d	4.38
C_Reviewed $\rightarrow$ Fixed	*16.3 d*	17.8 w	6.64

compared to Performance issues. Only 49.5% of the Performance issues are labeled with open status label during their lifecycle, even if labeled, the delay is more than Security issues (*BNR* is 2.51). Henceforth, it highlights the need to emphasize the importance of defined labeling practices in more efficient way for Performance issues.

3) We know that more percentage of Performance issues have patches for resolution. We observe that transitions from VCS to PCR and PCR to ITS is more efficient also as compared to Security issues. Delay to close PCR issue after successful commit is much higher for Security issues with *BNR* value 4.38. Similarly for marking an issue as *Fixed* after completion of patch review is taking too much time in case of Security issues (that is, *BNR* $= 6.64$).

B. General Bottleneck

Experimental results demonstrate transitions which take lot of time delaying an individual process. For example, following results are few actionable information for the project team:

1) We notice that the mean time to reopen and fix closed issues (*I_Closed* $\rightarrow$ *Fixed*) is very high, that is, 29.8 weeks for Security issues. At the same time this transition is present in many cases indeed, it emphasizes the need to minimize issue reopen.
2) In case of Security issues, it takes mean time of around 3 months to directly fix an issue that is, transition from *I_Creation* to *Fixed*. On the other hand, sequence of transitions as *I_Creation* $\rightarrow$ *I_Open* $\rightarrow$ *Fixed* takes mean time of around 2 months which indicates that proper assignment of issue before jumping to the resolution reduces overall delay.
3) Reopening of Performance issues, that is, *I_Closed* $\rightarrow$ *I_Open* takes mean time of 54.9 days which is undesirable and degrades the overall user perceived quality of the bug fixing process.
4) Resolution of an issue as *WontFix* or *Invalid* is taking significantly high mean time of around 21 and 17 weeks respectively for Performance issues. It indicates inefficiency in identifying the issues which are not worth fixing because of several reasons like insufficient information.
5) Control transfer from ITS state *I_Open* to PCR state *C_Creation* is taking 11.3 and 15 days for Security and Performance issues respectively. It indicates scope of improvement in making transition smooth between different IS.

VI. Conclusion

We model process cube with 9 dimensions for defect resolution process. We handle sparsity for selected dimension (D_{sel}) by meaningful clustering and hierarchies. We perform a set of OLAP operations (slice, dice and roll-up) to eventually compare Performance and Security issues. We derive two sublogs (each corresponding to a cell) from *EB* by materialization step. We obtain process model with 11 states which helps to visualize differences. Empirical analysis reveals interesting patterns like more Performance issues have patches, high percentage of Security issues are *Fixed*. We identify high number of unique traces for both *Types* with majority cases having 3 or 4 events in the lifecycle. We find exceptional control transfer in 18% of Performance issues. We observe frequent reopen cases (anti-patterns) for Security issues and presence of code review cycles (signature patterns). We identify *Exclusive* bottlenecks using *BNR* and also *General* bottlenecks delaying individual process.

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